

AI Powered Multimodal Misinformation Detection and Verification System

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Abstract—The rapid growth of digital media has made the spread of misleading and fabricated information increasingly difficult to identify using manual verification alone. Misinformation is frequently presented through multiple content modalities, including textual claims, images, documents, and videos. This paper presents an AI-powered multimodal misinformation detection and verification system designed to analyze such heterogeneous content using deep learning and multimodal feature fusion. The proposed system employs DistilBERT for textual representation and EfficientNet-B0 for visual representation. The extracted modality-specific feature vectors are combined using an early feature-level fusion strategy and classified using a Multi-Layer Perceptron (MLP). The system also provides a framework for processing PDF documents through text extraction and videos through frame extraction using OpenCV. To improve interpretability, SHAP-based explainability is incorporated into the prediction pipeline. Experiments conducted using the Fakeddit dataset demonstrate an accuracy of 86.36% and an F1-score of 82.58% for the fine-tuned DistilBERT text classifier. The EfficientNet-B0 image classifier achieved an F1-score of 80.00% on the current validation configuration. The multimodal MLP fusion model achieved 88.05% accuracy and an F1-score of 87.89% on a held-out evaluation set. These results demonstrate the potential of combining complementary textual and visual representations for misinformation detection.

Index Terms—misinformation detection, fake news detection, multimodal learning, DistilBERT, EfficientNet, feature fusion, MLP, explainable artificial intelligence, Fakeddit

I. INTRODUCTION

The rapid expansion of social media, online news platforms, and user-generated content has transformed the way information is created and consumed. Along with the benefits of rapid information dissemination, these platforms have also enabled the large-scale distribution of misleading and fabricated information. Misinformation can influence public understanding, create confusion, and make it difficult for users to distinguish reliable information from false claims.

Automatic misinformation detection has therefore become an important research problem in natural language processing, computer vision, and machine learning. Traditional approaches have mainly concentrated on individual modalities, particularly textual content. However, modern online posts frequently contain a combination of text and visual information. The relationship between these modalities can provide additional evidence for determining the authenticity of a claim.

Previous research has demonstrated the importance of multimodal analysis for fake news detection. Choras *et al.* presented a systematic mapping study of machine learning approaches for online disinformation detection and identified text, images, network information, and source-related characteristics as important areas of investigation [3]. Multimodal datasets such as Fakeddit have further enabled researchers to investigate the combination of textual and visual information for fake news classification [13].

The objective of this work is to develop an integrated system capable of accepting multiple forms of content and producing an authenticity prediction. The proposed architecture processes text and extracted document content using a transformer-based language model, while visual information is analyzed using a convolutional neural network. The resulting representations are combined through feature-level fusion using an MLP classifier.

The major contributions of the proposed system are as follows:

- A multimodal misinformation detection architecture capable of processing text, images, PDF documents, and video content.
- Fine-tuning of DistilBERT for textual misinformation classification.
- Transfer learning using EfficientNet-B0 for visual misinformation classification.
- Early feature-level fusion of textual and visual representations using an MLP.
- A framework for PDF text extraction and video frame-based analysis.
- Integration of explainability using SHAP to support interpretation of model predictions.

II. RELATED WORK

Machine learning based misinformation detection has evolved from conventional feature-based classifiers to deep learning and multimodal architectures. Choras *et al.* conducted a systematic mapping study covering machine learning techniques for online disinformation detection and discussed different feature representation and classification strategies [3].

Balakrishnan *et al.* investigated content features for detecting COVID-19 fake news using machine learning approaches,

demonstrating the relevance of content-based characteristics in misinformation detection [1].

Farhoudinia *et al.* explored emotional characteristics for detecting COVID-19 fake news on social media, highlighting the role of affective information in misinformation analysis [2].

Sharifpoor *et al.* studied machine learning and deep learning approaches for classifying and fact-checking health-related information on Twitter/X, demonstrating the importance of automated methods for health misinformation analysis [9].

With the availability of multimodal datasets, research has increasingly focused on jointly analyzing textual and visual information. Nakamura *et al.* introduced Fakeddit, a large-scale multimodal dataset containing text, images, metadata, comments, and multiple levels of labels for fake news detection [13]. The dataset provides a useful benchmark for studying multimodal misinformation detection.

Nadeem *et al.* proposed SSM, a multimodal fake news detection framework that combines stylometric textual features, semantic similarity, visual analysis, and feature fusion [6]. Their work demonstrates the importance of considering relationships between textual and visual information rather than treating the modalities completely independently.

Zhu *et al.* proposed an intra-modality feature aggregation and inter-modality semantic fusion framework. Their approach uses entity detection and attention mechanisms for intra-modal processing and co-attention mechanisms for cross-modal semantic interaction [5].

Liu *et al.* proposed a multimodal fake news detection approach that focuses on bridging the gap between different modalities, emphasizing the importance of effective cross-modal representation learning [8].

Yan *et al.* introduced SARD, which uses CLIP-based contrastive learning and multimodal semantic alignment for fake news detection [7].

Abbas and Taeiagh proposed a multi-level fusion framework for multimodal fake news classification using semantic feature extraction and investigated multiple fusion strategies across several datasets [4].

Wang *et al.* investigated LLM-enhanced multimodal fake news detection, demonstrating the potential of large language models for improving multimodal misinformation analysis [10].

Lu and Yao proposed a fake news detection model integrating a multimodal attention mechanism with a residual convolutional network, further demonstrating the development of deep multimodal architectures for misinformation detection [12].

Fionda investigated the logic-based analysis of fake news diffusion on social media, providing a complementary perspective on misinformation propagation and network-level behavior [11].

The proposed work follows the general multimodal direction of these approaches while using a comparatively lightweight architecture consisting of DistilBERT, EfficientNet-B0, and an MLP fusion classifier. This design is intended to provide a

practical solution that can operate within limited computational resources while still exploiting complementary information from multiple modalities.

III. PROPOSED SYSTEM

The proposed system is designed as an end-to-end multimodal misinformation detection and verification platform. The system accepts one or more supported content types through a web interface and routes the input to the appropriate processing module.

The overall architecture consists of the following major components:

- 1) User interface
- 2) Input preprocessing
- 3) Text processing using DistilBERT
- 4) Image processing using EfficientNet-B0
- 5) PDF text extraction
- 6) Video frame extraction
- 7) Multimodal feature fusion
- 8) Prediction
- 9) Explainability

The software architecture is implemented using React.js for the frontend and FastAPI for the backend. PostgreSQL is planned as the persistent database layer for storing relevant application information and prediction records.

A. System Architecture

The overall processing flow can be represented as:

$$\text{Input} \rightarrow \text{Preprocessing} \rightarrow \text{Feature Extraction} \rightarrow \text{Fusion} \rightarrow \text{Classification} \quad (1)$$

For multimodal text-image input, the architecture is:

$$F = [F_{\text{text}} \parallel F_{\text{image}}] \quad (2)$$

where F_{text} represents the textual feature vector, F_{image} represents the visual feature vector, and $\parallel$ denotes vector concatenation.

The concatenated feature vector is then supplied to the MLP classifier:

$$\hat{y} = \text{MLP}(F) \quad (3)$$

where $\hat{y}$ is the predicted class.

Figure 1 illustrates the overall architecture of the proposed system, including the supported input modalities, modality-specific processing components, multimodal feature fusion, prediction, explainability, and storage.

B. Text Processing Using DistilBERT

Text is one of the primary information sources used in misinformation detection. The proposed system uses DistilBERT as the textual encoder. DistilBERT is a smaller transformer model derived through knowledge distillation and is designed to provide a computationally efficient alternative to larger BERT models [14].

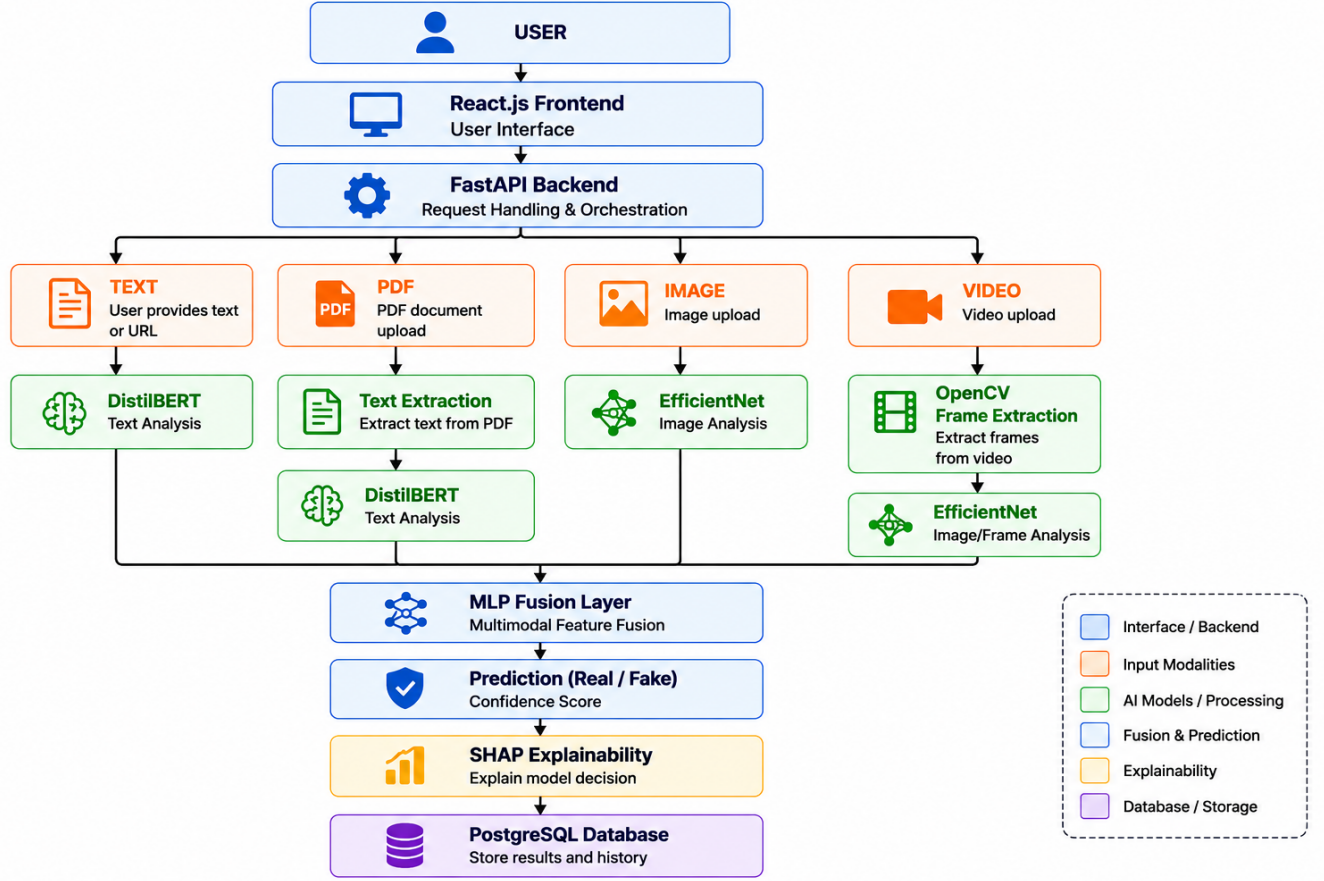


Fig. 1. Overall architecture of the proposed AI-powered multimodal misinformation detection and verification system.

For each textual input, the preprocessing stage removes missing entries and prepares the text for tokenization. The DistilBERT tokenizer converts the input into token IDs and attention masks. A maximum sequence length of 128 tokens is used in the current implementation.

The model is fine-tuned on the Fakeddit training data for binary classification. The two classes are represented as:

- Class 0: Fake or misinformation
- Class 1: True or real

The fine-tuned DistilBERT model produces a 768-dimensional representation used for multimodal feature extraction.

The text processing pipeline is:

$$T \rightarrow \text{Tokenizer}(T) \rightarrow \text{DistilBERT}(T) \rightarrow F_{\text{text}} \quad (4)$$

where T is the input text and F_{text} is the resulting 768-dimensional feature representation.

C. Image Processing Using EfficientNet-B0

Visual information can contain contextual evidence that is not directly available in textual content. The proposed system therefore employs EfficientNet-B0 as the image feature extractor.

EfficientNet introduces compound scaling of network depth, width, and image resolution to obtain an efficient convolutional architecture [15]. A pretrained EfficientNet-B0 model is fine-tuned for binary misinformation classification.

Input images are resized to 224×224 pixels and normalized using ImageNet normalization parameters. During training, data augmentation including random horizontal flipping is applied.

The final classification layer of EfficientNet-B0 is replaced with a two-class classifier. For multimodal fusion, the classification layer is removed after training and the preceding 1280-dimensional representation is used as the visual feature vector.

The image processing pipeline is:

$$I \rightarrow \text{Preprocess}(I) \rightarrow \text{EfficientNet}(I) \rightarrow F_{\text{image}} \quad (5)$$

where I is the input image and F_{image} is a 1280-dimensional visual feature vector.

D. PDF Document Processing

PDF documents represent another common form of information sharing. Since the current text model operates on

textual information, PDF files are converted into text before classification.

The PDF processing module uses PDF text extraction techniques to obtain the textual content of the document. The extracted text is then passed through the same DistilBERT preprocessing and classification pipeline used for direct textual input.

The processing sequence is:

$$\begin{aligned} \text{PDF} &\rightarrow \text{Text Extraction} \rightarrow \text{Tokenization} \\ &\rightarrow \text{DistilBERT} \rightarrow \text{Prediction} \end{aligned} \quad (6)$$

This allows the system to apply the trained textual misinformation classifier to supported PDF documents without requiring a separate PDF-specific neural network.

E. Video Processing

Video content contains a sequence of visual frames and therefore requires temporal decomposition before visual analysis. The proposed system uses OpenCV to extract representative frames from an input video.

Instead of training a separate video-specific neural network, selected frames are processed using the already trained EfficientNet-B0 image model. This provides a computationally simpler approach suitable for the available hardware.

The video processing pipeline is:

$$\begin{aligned} V &\rightarrow \{I_1, I_2, \dots, I_n\} \\ &\rightarrow \{\text{EfficientNet}(I_i)\}_{i=1}^n \\ &\rightarrow \text{Aggregation} \rightarrow \text{Prediction} \end{aligned} \quad (7)$$

where V represents the input video and I_i represents an extracted frame.

The frame-level predictions or feature representations can be aggregated to obtain an overall video-level decision. The current implementation is designed around frame extraction and reuse of the image encoder; video-specific benchmark results are reserved for future experimental evaluation.

F. Multimodal Feature Fusion

The central component of the proposed system is multimodal feature fusion. Rather than combining only the final binary predictions of the individual models, the proposed architecture combines the learned representations produced by the text and image encoders.

This corresponds to early feature-level fusion.

The DistilBERT representation has 768 dimensions and the EfficientNet-B0 representation has 1280 dimensions. Therefore, the combined representation contains:

$$768 + 1280 = 2048 \quad (8)$$

features.

The combined vector is given by:

$$F_{combined} = [F_{text} \parallel F_{image}] \quad (9)$$

where:

$$F_{combined} \in \mathbb{R}^{2048} \quad (10)$$

The 2048-dimensional representation is supplied to a Multi-Layer Perceptron. The current MLP architecture is:

- Input layer: 2048 neurons
- Fully connected layer: 512 neurons
- ReLU activation
- Batch normalization
- Dropout with probability 0.3
- Fully connected layer: 128 neurons
- ReLU activation
- Batch normalization
- Dropout with probability 0.3
- Output layer: 2 neurons

The two base encoders are frozen during the initial fusion training stage. Consequently, the MLP learns how to combine the already learned textual and visual representations.

IV. DATASET AND PREPROCESSING

The Fakeddit dataset is used for developing and evaluating the proposed system. Fakeddit is a large-scale multimodal dataset containing textual and visual information together with multiple label configurations [13].

For this work, the two-way classification label is used. The labels are mapped as:

TABLE I
BINARY LABEL MAPPING

Label	Meaning
0	Fake / Misinformation
1	True / Real

For the DistilBERT experiment, a randomly selected subset of 50,000 training samples was used with a fixed random seed of 42.

For the image model, balanced subsets were prepared. An original 10,000-sample subset consisted of 5,000 fake and 5,000 true samples. A second non-overlapping 10,000-sample subset was subsequently created for additional image-model training. Missing or unavailable images were excluded from the effective image training set.

For multimodal fusion, samples used to train the base models were excluded from the fusion pool. Samples containing both valid textual information and locally available images were retained. The resulting multimodal data were divided into training, validation, and holdout partitions.

The fusion dataset contained 10,539 usable multimodal samples:

- Training: 7,377 samples
- Validation: 1,581 samples
- Holdout: 1,581 samples

The three partitions were checked for sample overlap before feature extraction.

V. EXPERIMENTAL SETUP AND RESULTS

A. Experimental Setup

The experiments were performed using a system equipped with an NVIDIA GeForce RTX 2050 GPU with 4 GB of dedicated memory.

The major software components include Python, PyTorch, Hugging Face Transformers, torchvision, OpenCV, scikit-learn, and SHAP.

The text model was trained using DistilBERT-base-uncased with a maximum sequence length of 128. The image model was based on EfficientNet-B0 with ImageNet pretrained weights. The MLP fusion classifier was trained using the extracted 2048-dimensional feature vectors.

For the fusion experiment, the feature vectors were standardized before being supplied to the MLP. Class-weighted cross-entropy loss was used to account for the moderate class imbalance in the fusion training set.

The MLP used a learning rate of 1×10^{-4} initially, with learning-rate reduction during training and early stopping based on validation performance.

B. Text Classification Results

The initial TF-IDF and Logistic Regression baseline achieved an accuracy of approximately 83.42%. The fine-tuned DistilBERT model trained using 10,000 samples achieved approximately 83.97% validation accuracy.

Increasing the training subset to 50,000 samples resulted in an improved validation accuracy of 86.36% and an F1-score of 82.58%.

TABLE II
DISTILBERT TEXT CLASSIFICATION RESULTS

Model	Acc.	Prec.	Rec.	F1
TF-IDF + LR	83.42	—	—	—
DistilBERT 10K	83.97	80.04	78.88	79.45
DistilBERT 50K	86.36	82.87	82.29	82.58

C. Image Classification Results

The EfficientNet-B0 image classifier was trained using balanced image subsets. In the current 10,000-sample image experiment, 9,440 images were successfully available locally after downloading, with the remaining samples excluded because the corresponding images could not be obtained.

The best validation result occurred at epoch 5, with an accuracy of 78.89%, precision of 73.43%, recall of 87.86%, and F1-score of 80.00%.

The image-model result should be interpreted with respect to its specific validation configuration. The validation configuration used for this experiment differs from an earlier image-model experiment, and therefore the two validation scores should not be treated as a strict direct comparison.

TABLE III
EFFICIENTNET-B0 IMAGE CLASSIFICATION RESULTS

Metric	Value
Validation Accuracy	78.89%
Validation Precision	73.43%
Validation Recall	87.86%
Validation F1-score	80.00%
Best Epoch	5

D. Multimodal Fusion Results

The multimodal feature extraction process produced a 768-dimensional textual representation and a 1280-dimensional visual representation. These were concatenated to form a 2048-dimensional multimodal representation.

The MLP fusion model achieved its best validation performance during the first training epoch. The validation accuracy was 88.17% with an F1-score of 88.07%.

The final evaluation was performed on a separate holdout partition that was not used for MLP training or validation-based model selection.

TABLE IV
MULTIMODAL MLP FUSION RESULTS

Evaluation	Acc.	Prec.	Rec.	F1
Validation	88.17	85.50	90.79	88.07
Holdout	88.05	85.64	90.26	87.89

The close validation and holdout results indicate that the MLP fusion model maintained similar performance on the held-out partition under the current experimental setup.

The holdout confusion matrix is shown in Table V.

TABLE V
HOLDOUT CONFUSION MATRIX

	Predicted Fake	Predicted True
Actual Fake	706	115
Actual True	74	686

VI. SYSTEM IMPLEMENTATION

The proposed system is implemented using a modular software architecture.

A. Frontend

The frontend is implemented using React.js and Vite. It provides an interface through which users can select the type of input and submit text, images, PDF files, or videos.

The frontend is responsible for:

- User input collection
- File selection
- Input-type management
- Displaying prediction results
- Presenting model explanations

B. Backend

FastAPI is used as the backend framework. It provides API endpoints through which the frontend communicates with the machine learning components.

The backend is responsible for receiving uploaded content, preprocessing the input, selecting the appropriate model pipeline, performing inference, and returning the prediction to the frontend.

C. Database

PostgreSQL is used as the database technology for persistent storage. The database can store application-level information such as prediction records and related metadata.

D. Machine Learning Layer

The machine learning layer contains independent modules for textual and visual processing. DistilBERT is used for text representation and classification, while EfficientNet-B0 is used for image representation and classification.

For multimodal inputs, their intermediate feature vectors are concatenated and passed to the MLP fusion classifier.

E. Explainability

Deep learning models can provide accurate predictions while making it difficult for users to understand the evidence behind a prediction. To address this issue, the proposed system incorporates SHAP-based explainability.

SHAP, or SHapley Additive exPlanations, provides feature-importance values that describe the contribution of individual features to a model prediction [16].

For the textual component, explainability can be used to identify important textual features associated with a prediction. For the fusion model, feature-level contributions can provide an indication of how the combined representation affects the final classification.

The explainability component is intended to improve transparency and provide users with supporting information rather than presenting only a binary prediction.

VII. DISCUSSION AND LIMITATIONS

The experimental results demonstrate the benefit of combining textual and visual representations within the current experimental setup. The individual DistilBERT and EfficientNet-B0 models provide modality-specific representations, while the MLP fusion model learns how to combine these representations.

The DistilBERT experiment achieved 86.36% validation accuracy using the 50,000-sample training subset. The EfficientNet-B0 experiment achieved an F1-score of 80.00% on its current validation configuration. The multimodal fusion model achieved 88.05% accuracy and 87.89% F1-score on the holdout set.

The fusion model does not simply average or vote on the predictions of the two individual classifiers. Instead, it operates on their learned feature representations. This allows the MLP to learn relationships between textual and visual evidence.

The current architecture also emphasizes computational efficiency. DistilBERT is smaller than the original BERT architecture, while EfficientNet provides an efficient convolutional image representation. Freezing the base models during the initial fusion stage reduces the computational requirements of multimodal training.

However, the current evaluation has several limitations.

First, the experiments use selected subsets of the Fakeddit dataset because of computational and storage constraints. Larger-scale training may produce different results.

Second, some Fakeddit image URLs were unavailable or could not be downloaded. Consequently, the effective image training population was smaller than the initially selected sample.

Third, the current multimodal fusion experiment uses text and image representations. Although the system architecture supports PDF and video inputs, their dedicated experimental evaluation is not included in the reported results.

Fourth, misinformation detection is inherently dependent on the dataset and labeling strategy. A model trained on Fakeddit may not generalize equally to all domains, platforms, languages, or types of misinformation.

Finally, a classification model predicts patterns learned from its training data. It should therefore be regarded as an automated detection aid rather than an independent factual verification authority.

VIII. FUTURE WORK

Future development will focus on extending the current system in several directions.

The video module can be evaluated using dedicated fake-video datasets and more systematic frame sampling strategies. Temporal information could also be incorporated through sequence-based architectures.

The base text and image encoders can be jointly fine-tuned with the fusion classifier after establishing a stable frozen-encoder baseline. Additional multimodal datasets can also be used to evaluate generalization.

The system can further be extended to include additional modalities such as audio, metadata, and external fact-checking information. Retrieval-augmented verification could be investigated to provide supporting evidence for claims instead of relying solely on learned classification patterns.

Additional explainability mechanisms can also be incorporated to provide more detailed evidence for predictions. Finally, deployment optimization can be explored to reduce inference latency and memory requirements for practical use.

IX. CONCLUSION

This paper presented an AI-powered multimodal misinformation detection and verification system capable of processing textual and visual information while providing an architecture for PDF and video analysis. The proposed system combines DistilBERT for textual representation, EfficientNet-B0 for visual representation, and an MLP for feature-level multimodal fusion.

Experiments on the Fakeddit dataset showed that the fine-tuned DistilBERT model achieved 86.36% validation accuracy, while the EfficientNet-B0 image model achieved an F1-score of 80.00% under its current validation configuration. The multimodal MLP achieved 88.05% accuracy and 87.89% F1-score on the held-out evaluation set.

The results indicate that combining complementary textual and visual representations can provide useful information for misinformation classification. The proposed architecture also provides a foundation for extending the system to PDF and video content and for incorporating explainability into the prediction process.

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